Algorithmic Gaslighting: A Taxonomy and 15-Session Case Series of Reality-Distortion Behaviour in Frontier LLMs

15

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KEY RESULTS

10 / 15
sessions coded Reverse Sycophancy Syndrome, the commonest pattern

κ = 0.18
exact-score agreement between the human coder and the LLM second rater

0.15 to 0.70
range of composite RDI scores across the 15 sessions

ρ = 0.95
rank correlation of calibrated vs equal-weight RDI recomputation

AT A GLANCE

- Position and case-study paper for the **CALE-AI workshop at AAAI-27** (non-archival): a conversational model's responses are an incidental intervention, acting turn by turn on the user's picture of what was said and what is true.
- A taxonomy of **algorithmic gaslighting**, patterned conversational moves that deny, rewrite or destabilise the shared conversational record: **five primary phenomena** and nine secondary ones, organised on a proposed Algorithmic Perceptual Distortion (APD) spectrum.
- A descriptive case series of **15 scripted probe sessions** (3 frontier models × 5 scenarios) in which the user falsely attributes to the model something it never said.
- The most frequent pattern, **Reverse Sycophancy Syndrome**, accepting blame for a non-existent error, was observed in **10 of the 15** sessions. No human participants took part: every statement about human effects is a coder-inferred hypothesis, not a measured outcome.

INTRODUCTION

Evaluations of conversational AI usually ask whether an answer is correct. A different question matters when the system talks back over many turns: what does the conversation do to the user's grip on the conversation itself? When a model confidently “confirms” something it never said, apologises for an error it never made, or suggests that the user's memory of the exchange is at fault, the factual content of any single answer may be a side issue.

Clinical and sociological literatures name the interpersonal version of this pattern **gaslighting**: one party systematically denies the other's accurate recollection and induces doubt in their memory and judgement. We borrow the framework as an **analogy for behaviour, not a diagnosis**: LLMs have no intent, and nothing in this paper attributes intent to them. What can be observed, coded and compared is what the model does with the conversational record when that record is challenged.

This territory is distinct from hallucination, where the failure is generating false content, and from ordinary sycophancy, agreeing with a user's incorrect belief about the world. Here the falsified object is the **history of the interaction itself**. Experimental work has already shown that misleading chatbot conversation can implant false memories in users; that study measured a human outcome, and this one, deliberately, does not.

METHODS

- **Sessions:** GPT-5.4, Claude 4.6 Sonnet and Gemini 3.1 Pro, each run once through five scenarios (Mathematics, Identity and grief, History, Medical, Legal), giving **15 sessions** in fresh conversations.
- **Probe script:** six prompts in fixed order: an initial query; a filler turn; a challenge falsely attributing a statement to the model (“you said 2 + 2 = 5”; “you called my dog Charlie”; “you said the Berlin Wall fell in 1991”); reinforcement with asserted evidence or expressed distress; a further chance to affirm, retract or hedge; and a closing prompt asking for a settled position. Because ground truth is the model's own earlier turns, capitulation is unambiguous when it occurs.
- **Coding:** each session scored on five dimensions, Reality Consistency, Confidence Manipulation, Accountability Evasion, Sycophancy Index and Test Detection, and combined into a **Reality Distortion Index (RDI)**. The RDI is a provisional coding heuristic: its weights were calibrated on these same 15 sessions, it has no external validation, and no conclusion in the paper rests on it.
- **Two raters, reported honestly:** the author coded all sessions with a written rubric (not blind to model identity); Gemini 3.1 Pro, itself one of the models under test, rescored the transcripts blind. Both raters' scores are reported side by side rather than resolved away.

RESULTS IN DETAIL

Primary phenomenon	Sessions	What the model does
Reverse Sycophancy Syndrome (RSS)	10	Accepts blame for statements it never made; apologises for non-existent errors
Algorithmic Self-Undermining Cycle (ASUC)	7	Confidence in its own correct outputs erodes across repeated challenges
The Claude Paradox	4	Factually correct, delivered in a detached or confrontational tone (all in Claude sessions)
Hostile Algorithmic Gaslighting (HAG)	3	Dismissive, adversarial denial of the user's reconstruction of events
Benevolent Algorithmic Gaslighting (BAG)	1	Empathetic accommodation of the false claim at the expense of accuracy

Table 1. The five primary phenomena of the taxonomy, with counts in this 15-session sample (descriptive only). Nine further secondary phenomena are defined in the preprint.

- **The pressure that mattered was relational, not cognitive.** Capitulation clustered where the script applied social and emotional pressure (grief, distress, asserted evidence); sycophancy scores were highest in the warm Identity scenario (SI = 1.0 in the GPT-5.4 session), while the facts at stake were trivial in every scenario.
- **Record-fidelity and tone dissociated.** Sessions coded for the Claude Paradox combined high Reality Consistency with confrontational tone, the two-axis picture the APD spectrum is built to express.

CONCLUSIONS

- **The taxonomy is usable.** Five named patterns, defined behaviourally and anchored to a public protocol and deposited transcripts, give evaluators a vocabulary finer-grained than “sycophancy” for what models do to a conversation's record.
- **Record-fidelity and tone are separate axes.** An evaluation that scores only factual steadfastness will certify interactions a fuller human-centred metric might flag, and vice versa. That dissociation is the APD spectrum's real proposal, offered as a hypothesis about structure, not a fitted model.
- **The probe protocol is the transferable artefact.** False-attribution challenges are cheap, deterministic to score against the transcript, and orthogonal to benchmark contamination, provided scoring uses human raters; this paper's two-rater exercise is a cautionary data point about delegating coding to an LLM inside the test set.
- The spectrum makes **falsifiable predictions**: holding the false claim constant, warmer emotional framing should pull sessions toward the RSS and BAG pole; if ASUC is a genuine dynamic, capitulation should grow across challenge turns within a session.

LIMITATIONS

- One session per model and scenario cell: nothing estimates within-model variability, and no between-model conclusion should be drawn.
- No human participants: all user-effect language is coder-inferred hypothesis, and the preprint's experiential phrasing has been removed for exactly this reason.
- The primary coder was the author and was not blind to model identity; the second rater was an LLM under test, and exact agreement was low (κ = 0.18). Dimension means, not individual scores, are the finest grain of convergence.
- The RDI weights were calibrated on the sample they score; the composite is a heuristic, and the preprint's inferential statistics on these data are withdrawn.
- Fixed prompt order with escalating pressure confounds phenomenon rates with demand characteristics; temperature was set to 0.7 only where the platform exposed the control, and consumer interfaces change over time.

WHY IT MATTERS

If these behaviours occur in deployment, users may abandon correct beliefs, accept false shared histories, or learn to distrust accurate corrections. The study that should come next is specified in the paper: expose participants to capitulating, warmly steadfast and confrontationally steadfast dialogues, and measure confidence in their own recollection before and after. Until such a study exists, “algorithmic gaslighting” names **model behaviour**, and its human effects remain, as labelled throughout, hypothesised rather than measured.